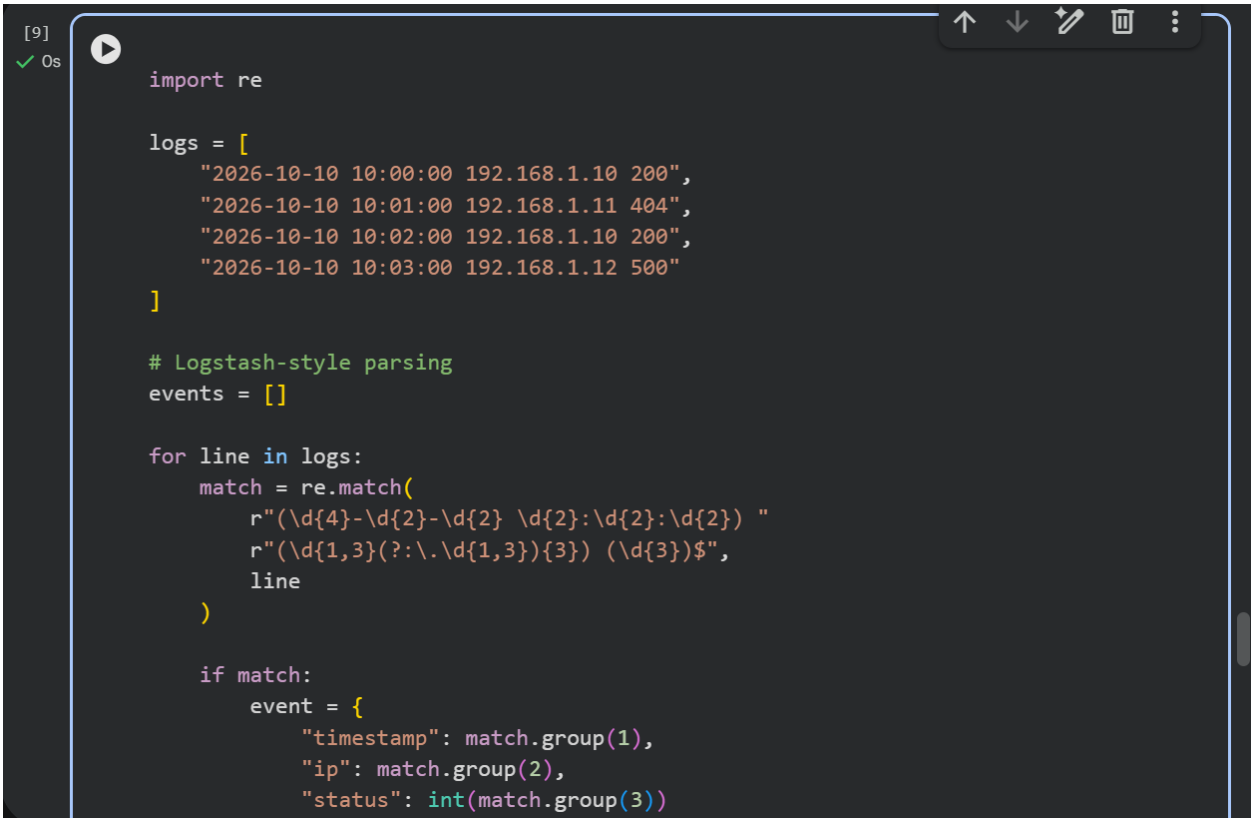
7. Write a Python program to reproduce the three ELK Stack roles: Logstash-style parsing of raw log lines into structured events, Elasticsearch-style indexing, and Kibana-style aggregation.



```
[9]
✓ 0s

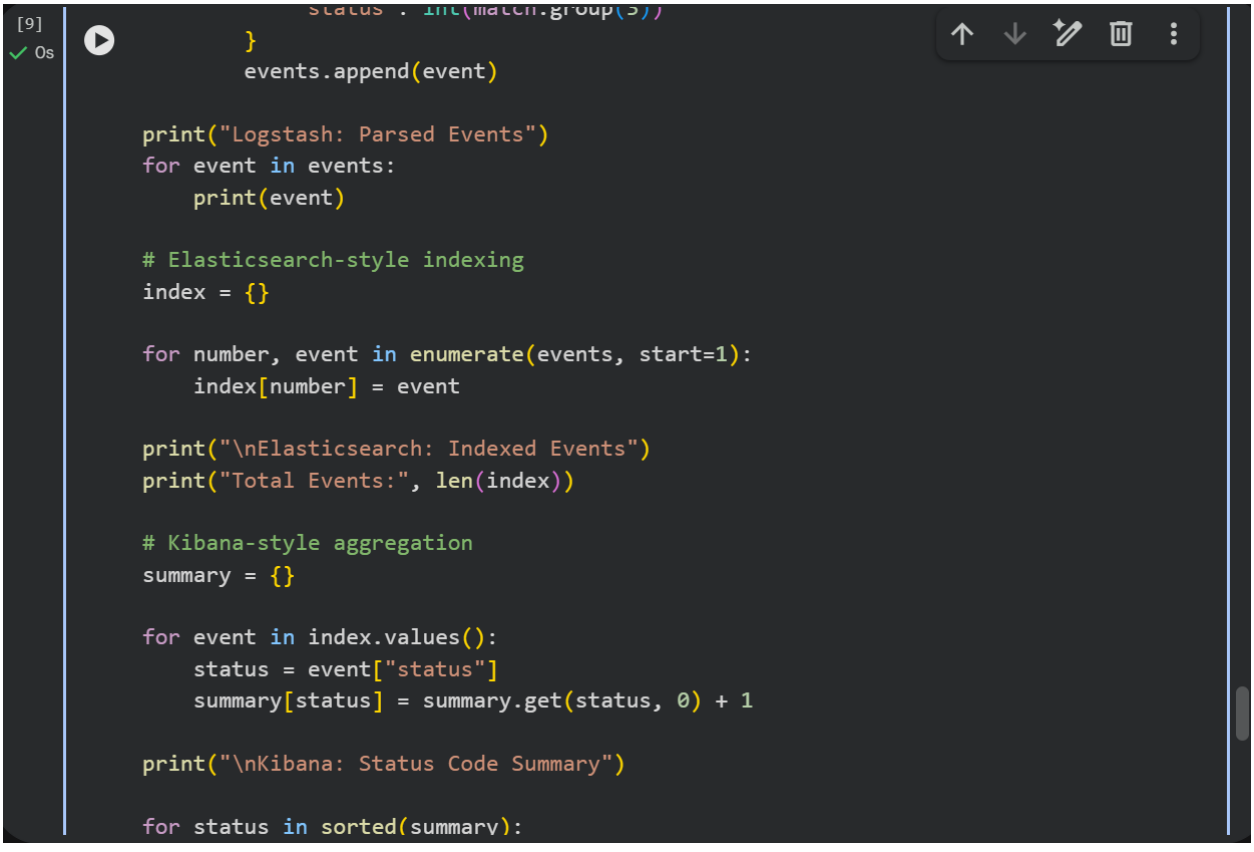
import re

logs = [
    "2026-10-10 10:00:00 192.168.1.10 200",
    "2026-10-10 10:01:00 192.168.1.11 404",
    "2026-10-10 10:02:00 192.168.1.10 200",
    "2026-10-10 10:03:00 192.168.1.12 500"
]

# Logstash-style parsing
events = []

for line in logs:
    match = re.match(
        r"(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}) "
        r"(\d{1,3}(?:\.\d{1,3}){3}) (\d{3})$",
        line
    )

    if match:
        event = {
            "timestamp": match.group(1),
            "ip": match.group(2),
            "status": int(match.group(3))
```



```
[9]
✓ 0s

            "status": int(match.group(3))
        }
        events.append(event)

print("Logstash: Parsed Events")
for event in events:
    print(event)

# Elasticsearch-style indexing
index = {}

for number, event in enumerate(events, start=1):
    index[number] = event

print("\nElasticsearch: Indexed Events")
print("Total Events:", len(index))

# Kibana-style aggregation
summary = {}

for event in index.values():
    status = event["status"]
    summary[status] = summary.get(status, 0) + 1

print("\nKibana: Status Code Summary")

for status in sorted(summary):
```

[9]
✓ 0s

```
        status = event["status"]
        summary[status] = summary.get(status, 0) + 1

    print("\nKibana: Status Code Summary")

    for status in sorted(summary):
        print(status, ":", summary[status])
```

```
Logstash: Parsed Events
{'timestamp': '2026-10-10 10:00:00', 'ip': '192.168.1.10', 'status': 200}
{'timestamp': '2026-10-10 10:01:00', 'ip': '192.168.1.11', 'status': 404}
{'timestamp': '2026-10-10 10:02:00', 'ip': '192.168.1.10', 'status': 200}
{'timestamp': '2026-10-10 10:03:00', 'ip': '192.168.1.12', 'status': 500}

Elasticsearch: Indexed Events
Total Events: 4

Kibana: Status Code Summary
200 : 2
404 : 1
500 : 1
```